

Chemical Kinetics

(NCERT BASED NOTES)

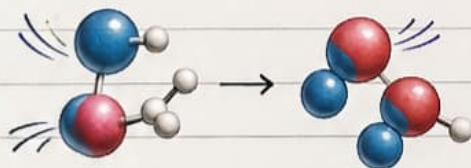
"The speed of a reaction decides the path of chemistry."

1. INTRODUCTION

Chemical Kinetics is the branch of chemistry that deals with the rate of chemical reactions and the factors affecting it. It helps us understand how fast a reaction occurs and through what mechanism.

KEY CONCEPTS

- Rate of reaction
- Factors affecting rate
- Rate law & rate constant
- Order and molecularity
- Activation energy
- Collision theory



Before Collision After Collision

Effective collision leads to product formation.

2. RATE OF REACTION

The rate of a reaction is defined as the **change in concentration** of reactants (or products) **per unit time**.

$$\text{Average rate of reaction} = - \frac{\Delta[\text{Reactant}]}{\Delta t}$$

(or) $\frac{\Delta[\text{Product}]}{\Delta t}$

$\Delta[\text{Reactant}]$ = decrease in concentration

$\Delta[\text{Product}]$ = increase in concentration

Δt = change in time

Example :

For the reaction $2A + B \longrightarrow \text{Products}$

$$\text{Rate} = - \frac{1}{2} \frac{\Delta[A]}{\Delta t} = - \frac{\Delta[B]}{\Delta t} = \frac{\Delta[\text{Product}]}{\Delta t}$$

Instantaneous rate :

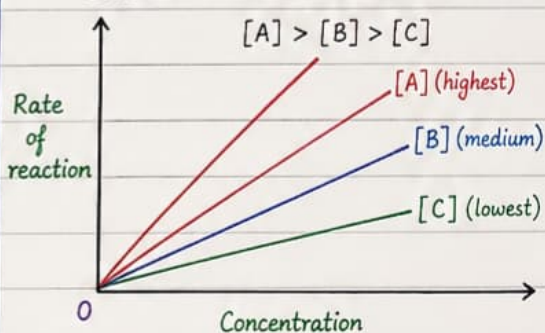
The rate at a particular instant of time is called instantaneous rate.

It is the slope of tangent to the curve at that point.

3. FACTORS AFFECTING RATE OF REACTION

- ① **Nature of Reactants** : Ionic compounds react faster than covalent compounds.
- ② **Concentration** : Higher the concentration, higher will be the rate.
- ③ **Temperature** : Increase in temperature increases the kinetic energy and rate.
- ④ **Catalyst** : A catalyst changes the rate without being consumed.
- ⑤ **Surface Area** : For solid reactants, greater the surface area, faster the reaction.

EFFECT OF CONCENTRATION ON RATE



REMEMBER THESE!

Rate depends on how frequently and how effectively particles collide.

More effective collisions
→ more products →
faster rate.

Catalysts provide an alternative path with lower activation energy.

4. ORDER OF REACTION

The order of a reaction is the **sum of the powers** to which the molar concentrations of the reactants are raised in the **rate law expression**.

Rate law for a general reaction



$$\text{Rate} = k[A]^x[B]^y \quad \begin{cases} x \rightarrow \text{order w.r.t. A} \\ y \rightarrow \text{order w.r.t. B} \end{cases}$$

Overall order = $x + y$

- Zero order reaction : Overall order = 0

$$\text{Rate} = k[A]^0 = k \text{ (constant)}$$

$$\text{Unit of } k = \text{mol L}^{-1} \text{ s}^{-1}$$

- First order reaction : Overall order = 1

$$\text{Rate} = k[A]^1 = k[A]$$

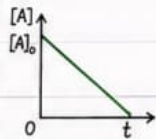
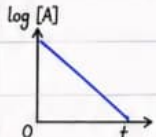
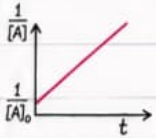
$$\text{Unit of } k = \text{s}^{-1}$$

- Second order reaction : Overall order = 2

$$\text{Rate} = k[A]^2 \quad \text{or} \quad k[A][B]$$

$$\text{Unit of } k = \text{L mol}^{-1} \text{ s}^{-1}$$

INTEGRATED RATE EQUATIONS

Order	Integrated rate equation	Graph	k (unit)
Zero order	$[A]_t = [A]_0 - kt$		$\text{mol L}^{-1} \text{ s}^{-1}$
First order	$\log [A]_t = \log [A]_0 - \frac{kt}{2.303}$		s^{-1}
Second order	$\frac{1}{[A]_t} = \frac{1}{[A]_0} + kt$		$\text{L mol}^{-1} \text{ s}^{-1}$

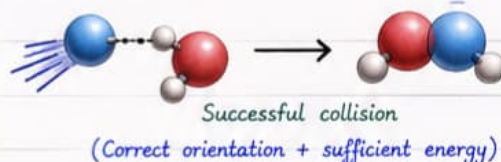


Important Points

- Order is determined experimentally.
- It can be zero, fractional or even negative.
- Overall order has no unit.
- Molecularity is always a whole number (1, 2 or 3).

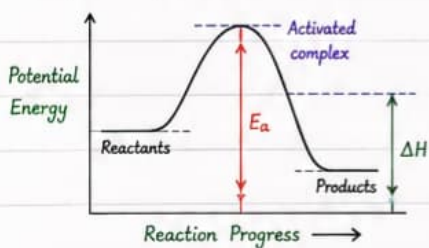
5. MOLECULARITY vs ORDER

Molecularity	Order
- Number of reacting species taking part in an elementary step. - Always a whole number (1, 2 or 3). - Determined from the balanced equation. - Has no experimental value.	- Sum of powers in the rate law expression. - Can be zero, fractional or negative. - Determined experimentally. - May or may not be same as molecularity.



6. ACTIVATION ENERGY (E_a)

It is the **minimum energy** that reactant molecules must possess for a reaction to occur.



Higher E_a
→ slower reaction
Lower E_a
→ faster reaction

7. COLLISION THEORY

A reaction occurs only when:

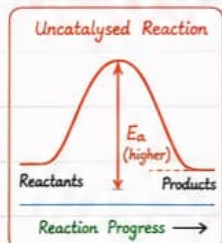
- Particles collide with each other.
- Collisions must have energy $\geq$ activation energy.
- Collisions must have proper orientation.

$$\text{Rate of reaction} \propto \frac{\text{Number of effective collisions}}{e^{E_a/RT}}$$

E_a = activation energy
 R = gas constant
 T = absolute temperature

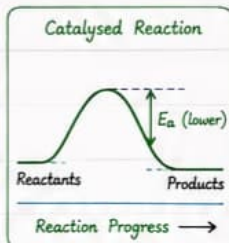
8. FACTORS AFFECTING ACTIVATION ENERGY

- Nature of reactants
- Presence of catalyst
- Solvent and medium
- Temperature



CATALYST

A catalyst provides an alternative pathway with lower activation energy and increases the rate of reaction without being used up.



QUICK REVISION

Rate depends on:

- Concentration
- Temperature
- Nature of reactants
- Catalyst
- Surface area

9. MECHANISM OF A REACTION

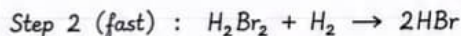
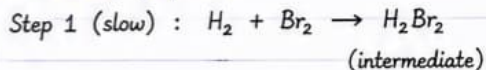
A complex reaction occurs in a **series of elementary steps**. The sequence of elementary steps by which overall reaction takes place is called the **mechanism**.

Key Points

- Intermediate** : Formed in one step and consumed in a subsequent step.
- Catalyst** : Takes part in a step but is regenerated.
- Rate-determining (slow) step** controls the overall rate.
- Fast steps are not seen experimentally.



Mechanism (proposed) :



Overall reaction is the sum of elementary steps.

Types of Elementary Steps

Type	Molecularity	Description	Example
Unimolecular	1	Involves one molecule	$\text{N}_2\text{O}_5 \rightarrow 2\text{NO}_2 + \frac{1}{2}\text{O}_2$
Bimolecular	2	Involves two molecules	$\text{Cl} + \text{NO}_2 \rightarrow \text{ClNO}_2$
Termolecular	3	Involves three molecules	$\text{O} + \text{O}_2 + \text{M} \rightarrow \text{O}_3 + \text{M}$

10. TEMPERATURE DEPENDENCE OF RATE

Increase in temperature **increases the rate** of reaction. This is due to increase in **kinetic energy** of molecules and more **effective collisions**.

Arrhenius Equation

$$k = A e^{-E_a/RT}$$

k = rate constant

A = frequency factor (pre-exponential factor)

E_a = activation energy

R = gas constant

T = absolute temperature

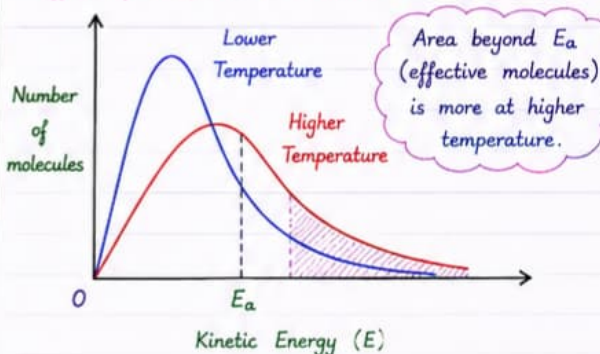
Higher T

→ more molecules have energy $\geq E_a$

→ more effective collisions

→ higher rate

Effect of Temperature on Maxwell Distribution

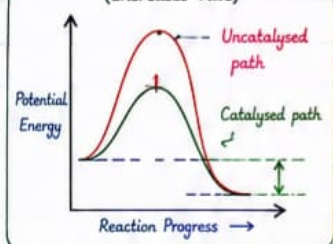


11. CATALYSIS

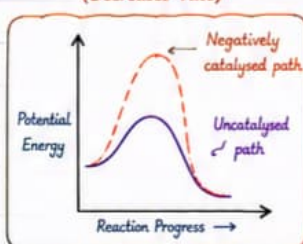
A catalyst increases the rate of reaction without being consumed. It provides an **alternative path** with lower activation energy.

Types of Catalysis :

(i) Positive Catalysis (Increases rate)



(ii) Negative Catalysis (Decreases rate)



Catalysts lower E_a → more effective collisions → faster reaction.

12. HALF-LIFE ($t_{1/2}$)

Half-life is the time required for the concentration of a reactant to become half of its initial value.

Order	Integrated Rate Equation	Half-life ($t_{1/2}$)
Zero order	$([\text{A}]_t = [\text{A}]_0 - kt)$	$t_{1/2} = \frac{[\text{A}]_0}{2k}$
First order	$(\log [\text{A}]_t = \log [\text{A}]_0 - \frac{kt}{2.303})$	$t_{1/2} = \frac{0.693}{k}$
Second order	$(\frac{1}{[\text{A}]_t} = \frac{1}{[\text{A}]_0} + kt)$	$t_{1/2} = \frac{1}{k[\text{A}]_0}$

♦ For zero order : $t_{1/2} \propto [\text{A}]_0$ (depends on initial concentration)

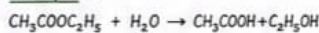
♦ For first order : $t_{1/2}$ is constant (independent of $[\text{A}]_0$)

♦ For second order : $t_{1/2} \propto \frac{1}{[\text{A}]_0}$ (inversely proportional)

13. PSEUDO-FIRST ORDER REACTIONS

When one reactant is in **large excess**, its concentration remains practically constant. The reaction then follows first order kinetics with respect to the other reactant.

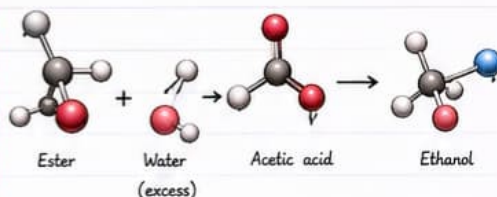
Example :



If $[\text{H}_2\text{O}]$ is in large excess,

$$\text{rate} = k' [\text{CH}_3\text{COOC}_2\text{H}_5]$$

where $k' = k [\text{H}_2\text{O}]$ (constant)



14. UNITS OF RATE CONSTANT (k)

From the rate law : $\text{Rate} = k[A]^x[B]^y$

Unit of rate = $\text{mol L}^{-1} \text{s}^{-1}$

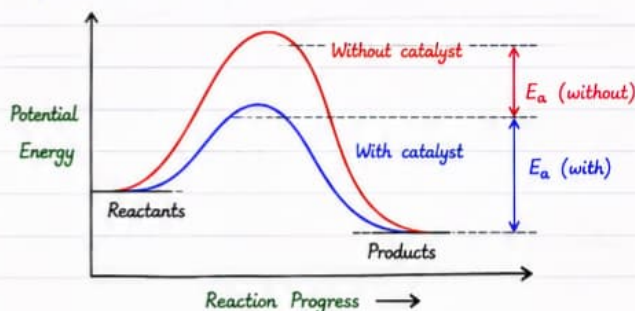
Order	Unit of k
Zero order	$\text{mol L}^{-1} \text{s}^{-1}$
First order	s^{-1}
Second order	$\text{L mol}^{-1} \text{s}^{-1}$
Third order	$\text{L}^2 \text{mol}^{-2} \text{s}^{-1}$
n^{th} order	$\text{L}^{n-1} \text{mol}^{1-n} \text{s}^{-1}$

★ IMPORTANT

- k depends on temperature and nature of reactants.
- For a given reaction, at a particular temperature, k is constant.

15. EFFECT OF CATALYST

A catalyst provides an alternative path with lower activation energy and increases the rate of reaction. It is not consumed in the reaction.



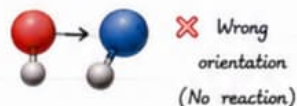
16. COLLISION THEORY (DETAILED)

- For a reaction to occur, particles must collide.
- But all collisions do not lead to reaction.

Effective collisions are those in which :

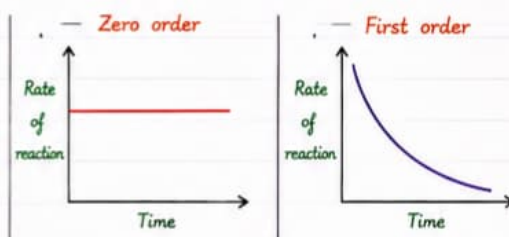
- Energy of collision $\geq$ activation energy
- Proper orientation of colliding molecules

ORIENTATION MATTERS



More effective collisions
→ more successful collisions
→ faster rate of reaction.

17. RATE vs TIME GRAPHS



KEY IDEA

As reaction proceeds, concentration of reactants decreases → fewer effective collisions → rate decreases (except in zero order).

18. NUMERICAL EXAMPLE (FIRST ORDER REACTION)

For a first order reaction, $t_{1/2} = \frac{0.693}{k}$

Example : The half-life of a reaction is 10 min. Calculate the rate constant.

Solution :

$$t_{1/2} = 10 \text{ min}$$

$$k = \frac{0.693}{t_{1/2}} = \frac{0.693}{10} = 0.0693 \text{ min}^{-1}$$

Answer : 0.0693 min^{-1}

ANOTHER FORMULA

For first order

$$k = \frac{2.303}{t} \log \frac{[A]_0}{[A]_t}$$

Where, $[A]_0$ = initial concentration

$[A]_t$ = concentration at time t

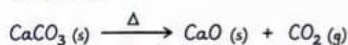
19. QUICK REVISION

- ✓ Rate depends on concentration, temperature, nature of reactants, catalyst.
- ✓ Order is found experimentally.
- ✓ k depends on temperature and nature.
- ✓ Higher activation energy → slower reaction.
- ✓ Catalyst provides an alternative path.
- ✓ Half-life is constant for first order reaction.

20. REAL LIFE APPLICATIONS

1. Decomposition of CaCO_3

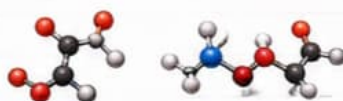
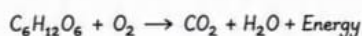
Limestone decomposes on heating in a kiln.



Rate depends on temperature.

2. Respiration in Humans

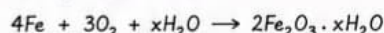
Glucose is oxidised in body cells slowly to release energy.



Enzyme acts as a catalyst and controls the rate.

3. Rusting of Iron

Iron reacts with oxygen and moisture slowly to form rust.



Rate increases in presence of moisture and decreases with protective coating.

SUMMARY

Reactants must collide with sufficient energy and proper orientation.

Activation energy is the minimum energy required for reaction.

Catalyst lowers activation energy and increases rate.

Rate laws and order help to predict reaction behaviour.

Chemical Kinetics explains how fast reactions occur and why.

Learn the concept, Practice problems & Believe in yourself - Success will follow!